



YMNOTES MEDICAL · DIGESTIVE SYSTEM

Additional Teaching Points

A chapter-by-chapter companion to the Digestive System study edition

Part 1 · Before the Mid-Module Exam · Chapters 1–15

Contains only the points in **GIT Internal Med** that are not already taught in the corresponding pages of the **YMnotes Digestive System** study edition (Version 3), book pages 1–197. Teach each chapter from the main book first, then use this companion for its additions.

BEFORE YOU START

How to use this companion

This is not a summary. Every entry is a point that the GIT Internal Med notes add to what the main book already teaches. Points that repeat or reword the main book have been left out on purpose.

Three kinds of entry

◆ ADDITION

A fact, mechanism, investigation detail or management point missing from the main book. Green rail, diamond marker.

● CLINICAL PEARL

A short, high-yield clinical clue or distinction. Blue double rail, round marker.

▲ DISCREPANCY · REVIEW

The two sources disagree. Both versions are shown with their page numbers, and neither is presented as correct. Check these before teaching. Red heavy rail, triangle marker.

Reading the page references

Reference	Means	PDF viewer page
Main p. 14	The page number printed on the YMnotes Digestive System book. Teach the entry at this point.	Printed page + 3 (p. 14 → PDF page 17)
IM p. 12	The page number printed on GIT Internal Med, where the point comes from.	The same as the printed page

Scope

- Chapters 1–15 of the main book, in its own order, up to printed page 197 (the last page of the supplied file). Chapter 15 therefore covers pages 195–197 only.
- Items are grouped under the main-book section where they should be taught, not in the order of the Internal Medicine notes.
- Chapters for which the Internal Medicine notes add nothing worthwhile still appear, with a short note, so the sequence matches the main book.

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Chapters

NO.	MAIN-BOOK CHAPTER · SUBJECTS	ENTRIES	PAGE
01	GI Symptoms and Investigations Internal Medicine · Radiology / Investigation · main book from p. 1	10 entries	3
02	The Esophagus Internal Medicine · Surgery · main book from p. 10	1 entry	6
03	The Stomach and Duodenum Internal Medicine · Pharmacology · Surgery · main book from p. 33	18 entries	7
04	Pathology of GIT Tumours Pathology · Surgery · main book from p. 70	none	13
05	Diarrhea and Malabsorption Internal Medicine · main book from p. 76	17 entries	14
06	Constipation Internal Medicine · main book from p. 95	7 entries	21
07	Foodborn Infection Microbiology · Internal Medicine · main book from p. 106	none	23
08	Inflammatory Bowel Disease Internal Medicine · Pharmacology · main book from p. 115	8 entries	24
09	Parasitic GI Diseases Parasitology · main book from p. 122	none	27
10	Small Intestine and Colon Surgery Surgery · main book from p. 140	none	27
11	Diverticulosis of the Colon Surgery · Internal Medicine · main book from p. 145	none	27
12	Appendix Surgery · main book from p. 151	none	27
13	Intestinal Obstruction Surgery · Emergency / Acute Care · main book from p. 162	none	27
14	Polyposis of Colon Surgery · Pathology · main book from p. 189	none	28
15	Colorectal Cancer Surgery · Pathology · main book from p. 195	none	28

61 entries in total: 40 additions, 12 clinical pearls, 9 discrepancies for review. Tap a chapter to jump to it.

CHAPTER 01 · TEACH AFTER MAIN BOOK FROM P. 1

01

GI Symptoms and Investigations

Internal Medicine · Radiology / Investigation 10 entries

Symptoms and signs Main pp. 2–5

● CLINICAL PEARL

Why alarm features matter

Many GI consultations are for functional (non-organic) symptoms, yet about **20% of all cancers** arise in the GI tract. The clinician's main task is therefore to separate organic disease from functional symptoms, and alarm features are the tool for doing so.

Teach with Main p. 2 · Alarm symptoms Source IM p. 1

C1-02

◆ ADDITION

Vomiting: how it is triggered

- ◆ **Vomiting centres** lie in the reticular formation of the medulla. They are stimulated by the **chemoreceptor trigger zone (CTZ)** and by the **vagus**; drugs, motion sickness and metabolic causes can stimulate these zones directly.
- ◆ Vomiting is usually preceded by two phases:
 - **Nausea**, with autonomic features (salivation, pallor, sweating).
 - **Retching**, a strong, involuntary, unproductive effort to vomit with contraction of the abdominal muscles.
- ◆ Add **vestibular disease** to the CNS causes of vomiting (alongside raised intracranial pressure and migraine).

Teach with Main p. 3 · Vomiting Source IM p. 2

C1-05/06/08

◆ ADDITION

Flatulence

Flatulence means excessive wind: belching, abdominal distension, gurgling, or passing flatus per rectum. Increased gas production comes from colonic bacterial breakdown of non-absorbed carbohydrate, a high-fibre diet, and carbonated drinks.

Teach with Main p. 3 · GI symptoms Source IM p. 2

C1-09

◆ ADDITION

Apparent acute abdomen: porphyria

Porphyria is another metabolic cause of an apparent "acute abdomen", alongside diabetic ketoacidosis.

Teach with Main p. 4 · Abdominal pain Source IM p. 4

C1-12

● CLINICAL PEARL

Pain patterns: diagnostic clues

- ◆ **Epigastric pain:** functional dyspepsia is the commonest diagnosis, but its symptoms can be **identical** to those of peptic ulcer disease.
- ◆ **Biliary pain** can be felt in the epigastrium as well as the right hypochondrium. It remits spontaneously and recurs weeks or months later.
- ◆ **Peptic ulcer** sometimes presents with right hypochondrial pain.
- ◆ Chronic, constant hypochondrial pain (right or left) in a **well-looking patient** is frequently functional. Right iliac fossa pain may also be functional.

Teach with **Main p. 4** · Abdominal pain by site Source **IM p. 4**

C1-13/14/15

● CLINICAL PEARL

Blood per rectum: where is it from?

- ◆ Dark blood and clots per rectum **without shock** always come from a lower GI source. An upper GI bleed produces blood per rectum only when it is massive, and then it is usually accompanied by shock.
- ◆ Haematochezia usually comes from a lesion distal to the **ligament of Treitz** (duodenojejunal junction). It can also occur with a massive upper GI bleed because of rapid transit.

Teach with **Main p. 4** · GI bleeding Source **IM pp. 5, 37**

C1-19/20

Investigations Main pp. 5–9

◆ ADDITION

Endoscopy: practical details

- ◆ **Sigmoidoscopy** is also used for diarrhoea and change in bowel habit. The **rigid** scope inspects at most 20–25 cm of distal colon; the **flexible** scope can reach the splenic flexure.
- ◆ **Colonoscopy:** benign strictures are **dilated**; malignant strictures are **stented**.
- ◆ **Balloon enteroscopy** (single or double balloon) can examine the small bowel from the duodenum to the ileum, using specialised enteroscopes in expert centres.
- ◆ **Capsule endoscopy:** avoid it if a stricture is suspected.

Teach with **Main pp. 6–7** · Endoscopy Source **IM pp. 6–7**

C1-23/24/25/26

◆ ADDITION

Imaging: extra interpretation points

- ◆ **Plain films:** chest as well as abdominal films are used in the acute abdomen; interpretation depends on gas shadows inside and outside the bowel. Plain films can also assess **faecal loading** in constipation. CT is more sensitive than plain films for gallstone and chronic pancreatitis calcification.
- ◆ **Ultrasound** in the acute abdomen can also show enlarged mesenteric glands.
- ◆ **CT:** oral intraluminal contrast (e.g. Gastrografin) with IV contrast improves diagnosis. CT is sensitive to **small volumes of gas** from a perforated viscus and to **contrast leak** from the gut lumen. In **high-grade** obstruction it is usually diagnostic of both the presence and the cause. It can guide biopsy of tumours or lymph nodes.
- ◆ **PET** is also used to detect **recurrent** disease.

Teach with Main pp. 7–9 · Imaging Source IM pp. 7–8

C1-29/30/31/34

◆ ADDITION

CT colonography: how it is done

CT colonography ("virtual colonoscopy") is performed after CO₂ insufflation into a previously cleansed colon. It is an alternative to colonoscopy for diagnosing colonic mass lesions.

Teach with Main p. 9 · CT colonography Source IM p. 8

C1-32

◆ ADDITION

Contrast studies

- ◆ Gastroscopy replaced the barium meal because it is **more sensitive** and allows biopsy.
- ◆ **Small bowel follow-through** shows the fold pattern and calibre of the small bowel; dedicated views of the terminal ileum show **early changes of Crohn's disease or TB**.
- ◆ **Small bowel enema (enteroclysis):** a tube is passed into the duodenum and a large volume of dilute barium is introduced, to show strictures or adhesions when intermittent obstruction is suspected. It has generally been replaced by **MR enteroclysis**.

Teach with Main p. 9 · Contrast studies Source IM p. 9

C1-35/36/37

The Esophagus

Internal Medicine · Surgery 1 entry

Your book already reproduces almost all of the oesophagus material in GIT Internal Med (IM pp. 10–20), including GERD, Barrett's oesophagus, hiatus hernia and achalasia. Only one addition was found.

Hiatus hernia: surgery Main p. 20

◆ ADDITION

Two further indications for surgery

Besides failure of medical treatment, IM lists two further indications for anti-reflux / hiatus hernia repair:

- ◆ a large hiatus hernia;
- ◆ **complications:** ulceration, severe inflammation, bleeding, stricture or Barrett's oesophagus.

Teach with Main p. 20 · Indications for surgery Source IM p. 17

C2-14

CHAPTER 03 · TEACH AFTER MAIN BOOK FROM P. 33

03

The Stomach and Duodenum

Internal Medicine · Pharmacology · Surgery 18 entries

Anatomy, physiology and gastritis Main pp. 34–36

◆ ADDITION

Gastric mucus and the functions of acid

- ◆ Mucus-secreting cells are present **throughout the stomach** and secrete both mucus and bicarbonate. The mucus is made of glycoproteins called **mucins**.
- ◆ Gastric acid is **not essential for digestion**, but it protects against some food-borne infections.
- ◆ Other gastric functions: a **reservoir** for food, **emulsification of fat** and mixing of gastric contents.

Teach with Main p. 35 · Anatomy and physiology Source IM pp. 21–22

C3-02/03

◆ ADDITION

Classifying gastritis and its causes

- ◆ Gastritis may be classified by **acuity** (acute vs chronic), by **histology**, or by **etiology**.
- ◆ **Gastropathy** (epithelial injury and regeneration with little inflammation) is usually caused by **bile reflux**, **hypovolaemia**, or **chronic congestion** as in portal hypertension.
- ◆ Named causes of gastritis:
 - H. pylori gastritis, the **commonest** cause
 - erosive and haemorrhagic gastritis (NSAIDs)
 - autoimmune gastritis
 - viral infection (CMV, herpes simplex)
 - Crohn's disease

Teach with Main pp. 35–36 · Gastritis Source IM p. 23

C3-05/06

◆ ADDITION

Acute gastritis: extra points

- ◆ Stress gastritis also occurs in **acute myocardial infarction** and **stroke**, as well as after surgery, trauma, burns and CNS injury.
- ◆ Most acute gastritis settles within days to weeks, but some cases become recurrent or long-term, i.e. **chronic gastritis**. Over time this raises the risk of gastric ulcer, upper GI bleeding and some gastric neoplasms.
- ◆ Treatment: an **H2 blocker** (e.g. famotidine) may be used along with the PPI. Remove risk factors: stop NSAIDs, reduce caffeine, stop alcohol and smoking.

Teach with **Main p. 36** · Acute gastritis Source **IM pp. 24-25**

C3-08/09/11

▲ DISCREPANCY · REVIEW

Autoimmune gastritis: "pangastritis"?

The two sources use the term "pangastritis" differently.

Main book	pp. 38, 42 say autoimmune gastritis affects the fundus and body and sparing the antrum ; "pangastritis" is used only for H. pylori gastritis spreading from antrum to body.
GIT Internal Med	p. 26 describes autoimmune gastritis as affecting the body and fundus and calls it "pangastritis".

Teach with **Main pp. 38, 42** · Chronic autoimmune gastritis Source **IM p. 26**

C3-13

● CLINICAL PEARL

Where intestinal metaplasia comes from

Gastric metaplasia, usually of the **intestinal type**, occurs almost always in the setting of **atrophic gastritis**. (Your book's Chapter 4, p. 72, states that intestinal-type gastric cancer arises on intestinal metaplasia; this entry supplies the missing first link.)

Teach with **Main pp. 38, 42** · Chronic gastritis Source **IM p. 26**

C3-14

Helicobacter pylori Main pp. 39–41, 48

▲ DISCREPANCY · REVIEW

Triple therapy: duration and first-line status

Main book	p. 41 gives PPI + amoxicillin + clarithromycin for 14 days but says it is no longer recommended as empirical first-line unless local clarithromycin resistance is known to be low; p. 48 gives 10–14 days .
GIT Internal Med	p. 31 presents the same triple therapy for 14 days as "one of the most commonly used" regimens.
For review	The main book is internally inconsistent on duration (14 vs 10–14 days), and the two sources differ on whether triple therapy is still first-line.

Teach with Main pp. 41, 48 · Eradication therapy Source IM p. 31

C3-17

● CLINICAL PEARL

Why combination therapy, and who is treated

- ◆ No single drug is effective against *H. pylori* on its own, so regimens combine two antibiotics with powerful acid suppression.
- ◆ Resistance to metronidazole and clarithromycin is high in some populations, and oral metronidazole frequently causes side-effects.
- ◆ All patients with duodenal or gastric ulcer who have *H. pylori* should receive eradication therapy. Treating incidental infection with **no ulcer** is controversial.

Teach with Main pp. 40–41, 48 · Eradication therapy Source IM p. 31

C3-18/19

Erosive gastritis and NSAIDs Main pp. 36–37, 48

● CLINICAL PEARL

NSAID damage: how common, and why it can be missed

- ◆ Aspirin inhibits the synthesis of both prostaglandin E2 and F2-alpha.
- ◆ With recurrent NSAID use (e.g. osteoarthritis) erosive gastritis can become **chronic**, and in alcoholics and chronic NSAID users it may be **completely asymptomatic**.
- ◆ Only about 5% of NSAID users have GI symptoms and 1–2% a major problem such as a GI bleed. It is still a large problem because so many people, especially the elderly, take NSAIDs, including low-dose aspirin.

Teach with Main pp. 36–37 · Gastritis and NSAIDs Source IM pp. 32, 35

C3-21/27

◆ ADDITION

Drug options for erosive gastritis

IM lists three drug options for erosive and haemorrhagic gastritis: **H2 blockers, proton pump inhibitors and sucralfate**. In the main book, sucralfate appears only in the pharmacology table (mainly stress-ulcer prophylaxis).

Teach with **Main p. 48** · Pharmacology Source **IM p. 32**

C3-23

Acute upper GI bleeding Main pp. 50–52

◆ ADDITION

Drugs and bleeding: two qualifiers

- ◆ **Even 75 mg/day of aspirin** can produce ulcers, erosions and GI haemorrhage.
- ◆ IM states that corticosteroids in the usual therapeutic doses have **no influence on GI haemorrhage** on their own. (Main p. 50 still advises avoiding NSAID–corticosteroid combinations.)

Teach with **Main p. 50** · Etiology Source **IM p. 37**

GB-03

◆ ADDITION

Natural history and prognosis

- ◆ Any significant GI bleed within the last **48 hours** should be assessed in hospital.
- ◆ About **85%** of patients stop bleeding spontaneously within 48 hours; many need no immediate treatment because the blood loss is small.
- ◆ Mortality from GI haemorrhage has stayed at **5–12%** despite advances in management, mainly because patients are now older and have more co-morbidity.

Teach with **Main p. 50** · Clinical approach Source **IM pp. 37–38, 48**

GB-04/31

◆ ADDITION

Resuscitation and history: practical triggers

- ◆ **Circulation:** unless there are clear signs of a cardiac cause, give IV fluid to any patient with **cool peripheries and a fast heart rate**.
- ◆ **Disability:** common causes of unconsciousness are profound hypoxia, hypercapnia, cerebral hypoperfusion, or recently given sedatives/analgesics.
- ◆ **History:** co-morbidity; acid-peptic symptoms; alcoholic liver disease, chronic hepatitis or cirrhosis; anticoagulants, antiplatelets, NSAIDs, steroids, alcohol binge; coagulation disorders. Ask about **epistaxis or haemoptysis** to exclude a non-GI source of the blood.

Teach with **Main p. 50** · ABCDE; unstable patient Source **IM p. 38**

GB-05/06

◆ ADDITION

Transfusion and supportive care

- ◆ Monitor pulse and BP **half-hourly**; cross-match **2 units** initially. Give plasma expanders or 0.9% saline until blood is available, then restore volume rapidly, preferably with red cell concentrates through large-bore cannulae.
- ◆ **Transfuse if:** shock (pallor, cold nose, systolic BP < 100 mmHg, pulse > 100/min), or Hb < 10 (IM prints the unit as "mg/dl") with recent or active bleeding.
- ◆ Hb falls late because **haemodilution has not yet occurred**. Pulse rate and venous pressure are the best guides to adequate transfusion. Insert a **CVP line** if there is organ failure or a high risk of heart failure, and watch for fluid overload, especially in the elderly.
- ◆ A **nasogastric tube** allows gastric wash and monitoring of ongoing bleeding.
- ◆ Clopidogrel and warfarin are also stopped. Stopping antiplatelets needs urgent discussion with a **cardiologist**. Manage patients in **high-dependency beds**.

Teach with Main pp. 50–51 · Unstable patient Source IM pp. 39–40

GB-07/08/09/11

● CLINICAL PEARL

Stigmata of recent haemorrhage

- ◆ Endoscopy finds the cause in **80% or more** of cases.
- ◆ Stigmata of recent bleeding are a **spurting vessel, active oozing, fresh or organised clot, or black spots**. Their presence means a **higher risk of rebleeding**, which is why they are an indication for endoscopic haemostasis.

Teach with Main p. 51 · Endoscopy Source IM p. 40

GB-12

◆ ADDITION

Endoscopic haemostasis: how each method works

- ◆ **Why combine:** dual or triple therapy reduces rebleeding more than monotherapy.
- ◆ **Adrenaline injection** gives temporary local tamponade and vasoconstriction: 0.5–1.0 mL aliquots of epinephrine 1:10,000 into four quadrants of the ulcer, within 1–2 mm of the bleeding point. Used **alone** it is less effective than other monotherapies. Its main use in active bleeding is **pre-injection**, to slow bleeding and improve the view before definitive therapy. Sclerosants (e.g. ethanolamine) and thrombogenic agents are less efficacious.
- ◆ **Thermal:** contact probes compress the vessel and then coagulate it (coaptive coagulation); the **bipolar probe** is the most widely used. **Argon plasma coagulation** is non-contact: monopolar current delivered through ionised argon gas. Other options are the heater probe and laser.
- ◆ **Clips** are the therapy of choice for **obvious arterial bleeding**. Unlike cautery they cause no further tissue damage, and they reduce rebleeding.

Teach with Main p. 51 · Endoscopic treatment Source IM pp. 40–42

GB-13/14/15/16

◆ ADDITION

Specific causes and after-care

- ◆ **Mallory-Weiss tear:** caused by a sudden rise in intra-abdominal pressure. A history of retching **may be absent**. Most bleeds are minor and patients are usually discharged within 24 hours, though the haemorrhage can occasionally be large.
- ◆ **After a bleeding ulcer:** continue the PPI for **4 weeks**; always **confirm H. pylori eradication**; if eradication fails, give long-term acid suppression.
- ◆ **Bleeding gastric cancer:** bleeds are usually not large, but oozing from the tumour is very difficult to control endoscopically. Surgery is occasionally needed for uncontrolled or repeated bleeding and can usually wait until full evaluation. Radiotherapy is occasionally successful.
- ◆ **After PCI:** heparins and clopidogrel increase GI bleeding. Give an IV PPI bolus followed by an infusion; platelet infusion can counter clopidogrel. Stopping antiplatelets carries a high risk.

Teach with **Main pp. 50–52** · Causes; peptic ulcer bleeding Source **IM p. 45**

GB-19/20/21/22

Acute lower and chronic GI bleeding Main pp. 52–53

◆ ADDITION

Lower GI bleeding: sorting the source

- ◆ **Unstable patient:** perform **nasogastric lavage**. Blood in the aspirate identifies an upper GI source; a clear aspirate makes a lower GI source more likely. Then proceed to angiography with embolisation, endoscopic intervention or, in extreme cases, surgery. Each of these both locates and treats the bleeding.
- ◆ **Stable patient:** abdominal and pelvic CT with oral contrast is used to look for the cause.
- ◆ **Angiography** has a low yield, so it is a test of last resort.

Teach with **Main pp. 52–53** · Acute lower GI bleeding Source **IM pp. 46–47**

GB-25/26/27

● CLINICAL PEARL

Chronic GI blood loss

- ◆ The cancers to exclude are particularly **stomach** and **right colon**.
- ◆ **Occult blood tests** are **unhelpful** in this work-up.
- ◆ Oesophageal varices usually bleed overtly and **rarely** present as chronic blood loss. **Hookworm** is the most common cause worldwide.
- ◆ Do upper endoscopy and colonoscopy **in the same session**. A colonic polyp found at colonoscopy should **not** be assumed to be the cause.
- ◆ In frail patients, **unprepared CT** is a reasonable test for colon cancer. Capsule endoscopy yields a diagnosis in **60–85%** but cannot treat.

Teach with **Main p. 53** · Chronic GI bleeding Source **IM p. 48**

GB-28/29

04

CHAPTER 04 · TEACH AFTER MAIN BOOK FROM P. 70

Pathology of GIT Tumours

Pathology · Surgery 0 entries

No substantive additions identified. GIT Internal Med has no tumour-pathology section. Its incidental mentions of GI tumours (e.g. MALT lymphoma with H. pylori, colon cancer as an alarm diagnosis) are either already in your book or listed under Chapters 1 and 3.

Diarrhea and Malabsorption

Internal Medicine 17 entries

Physiology of fluid and nutrient handling Main pp. 77–78, 86

◆ ADDITION

Where intestinal water comes from and why it moves

- ◆ Water crosses the epithelium **passively**. It follows osmotic gradients created by **active transport** of electrolytes (Na^+ , Cl^-) and of other solutes such as sugars and amino acids.
- ◆ Most of the ~9 L delivered to the small intestine each day is **endogenous secretion**, not oral intake:

Daily fluid delivered to the small intestine (IM p. 49)

Source	Volume (mL)
Oral intake	2,000
Salivary secretion	1,500
Gastric secretion	2,500
Pancreatic secretion	1,500
Biliary secretion	500
Small intestinal secretion	1,000
Total	9,000

Teach with Main pp. 77–78 · Basic physiology Source IM p. 49

C5-02/03

◆ ADDITION

Digestive enzymes by source

Amylase breaks starch into oligosaccharides, and brush-border enzymes finish the job: lactase splits lactose into glucose + galactose, and sucrase splits sucrose into glucose + fructose.

Digestive enzymes by source (IM p. 50)

Source	Enzymes
Salivary glands	Amylase, lingual lipase
Stomach	Pepsin
Pancreas	Amylase, trypsin, chymotrypsin, carboxypeptidase, elastase, lipase–colipase, phospholipase A2, cholesterol esterase
Intestinal mucosa	Enterokinase; disaccharidases (sucrase, maltase, trehalase, α -dextrinase/isomaltase); peptidases (aminooligopeptidase, dipeptidase)

Teach with Main p. 78 · Digestion and absorption (Fig 5.1) Source IM p. 50

C5-04

◆ ADDITION

Where nutrients are absorbed

The site of disease predicts the deficiency:

- ◆ **Duodenum and jejunum:** calcium, iron, magnesium, phosphorus, folate, the fat-soluble vitamins (A, D, E, K), and lipids, monosaccharides and small peptides.
- ◆ **Ileum:** vitamin B12 and bile salts and acids (also vitamin C, folate, vitamins D and K, magnesium). Many other nutrients can be absorbed here depending on transit time.
- ◆ **Large intestine:** water, sodium, chloride, potassium, short-chain fatty acids, vitamin K and biotin.

Teach with Main p. 86 · Malabsorption: clinical consequences Source IM p. 62

C5-26

Mechanisms and causes of diarrhoea Main pp. 77–84

▲ DISCREPANCY · REVIEW

How long is "acute" diarrhoea?

Main book	p. 77: acute diarrhoea lasts less than 2 to 3 weeks or, rarely, 6 to 8 weeks.
GIT Internal Med	p. 51: acute diarrhoea lasts less than 4 weeks .
For review	Both sources agree that diarrhoea lasting more than 4 weeks warrants evaluation as chronic.

Teach with Main p. 77 · Definition Source IM p. 51

C5-08

◆ ADDITION

Osmotic and secretory diarrhoea: the mechanism behind the lists

- ◆ **Osmotic:** usually a poorly absorbable carbohydrate or a **divalent ion** (e.g. magnesium). The high luminal osmolality draws water into the duodenum and jejunum to make chyme isotonic. Na^+ then follows the water into the lumen down its concentration gradient, and this Na^+ pulls in further water.
- ◆ **Secretory laxatives** (examples for your book's "laxatives"): ricinoleic acid (castor oil) and dioctyl sodium sulfosuccinate.

Teach with Main pp. 79–80 · Classification of diarrhoea Source IM pp. 52–53

C5-09/12

● CLINICAL PEARL

Why oral rehydration works even in cholera

Even while cholera toxin (via cAMP) drives chloride secretion from the crypts, villus cells can still absorb sodium, and with it water and chloride, through **coupled uptake of sodium with glucose or amino acids**. Oral rehydration solution uses this intact pathway.

Teach with Main p. 82 · Treatment of acute diarrhoea Source IM p. 56

C5-17

◆ ADDITION

Acute infectious diarrhoea: clues from the exposure

Vehicle and classic pathogens (IM p. 55)

Vehicle	Classic pathogens
Water	Vibrio cholerae, Giardia
Poultry	Salmonella
Seafood and shellfish (e.g. sushi)	Salmonella and Shigella
Canned vegetables or fruits	Clostridium species
Cheese, milk	Listeria species
Hospitalisation, antibiotics or chemotherapy	C. difficile
Animal-to-person (pets)	Salmonella, Campylobacter, Cryptosporidium

Teach with Main pp. 81–82 · Acute diarrhoea Source IM p. 55

C5-15

◆ ADDITION

Clostridium difficile colitis

Your book names *C. difficile* only as a cause. IM adds:

- ◆ **Pathogenesis:** antibiotics disrupt the normal flora; *C. difficile* colonises the colon and releases **toxins** that cause mucosal damage and inflammation (originally called pseudomembranous colitis).
- ◆ **Diagnosis:** diarrhoea (≥ 3 unformed stools in 24 h) **or** radiographic ileus/toxic megacolon, **plus** a positive stool *C. difficile* toxin test **or** pseudomembranes at colonoscopy/pathology.
- ◆ **Treatment:** stop the offending antibiotic. **Metronidazole** is the drug of choice for mild infection; **vancomycin** for more severe episodes; **fidaxomicin** (a macrocyclic antibiotic) is more active than vancomycin.
- ◆ Probiotics are promising for prevention and treatment. **Faecal microbiota transplantation** is reserved for refractory *C. difficile* colitis. Relapse is common (five or more relapses have been observed).

Teach with **Main p. 83** · Refractory diarrhoea Source **IM pp. 57–58**

C5-19

◆ ADDITION

Post-infectious IBS: treatable causes to look for

- ◆ Some cases are due to **bacterial overgrowth** and may respond to non-absorbable antibiotics; others are due to **acquired bile acid malabsorption**.
- ◆ **Coeliac disease** can first appear after an intestinal infection, so screen for it in this setting.

Teach with **Main p. 84** · Post-infectious IBS Source **IM p. 58**

C5-21

◆ ADDITION

Chronic diarrhoea: think of the patient group

Chronic diarrhoea in specific patient groups (IM p. 59)

Patient group	Causes to consider
Travellers	Bacterial infection (mostly acute); protozoal (amoebiasis, giardiasis); tropical sprue
Epidemics and outbreaks	Bacterial infection; epidemic idiopathic secretory diarrhoea (e.g. Brainerd diarrhoea); protozoal (cryptosporidiosis); viral (rotavirus, COVID-19)
Diabetes mellitus	Altered motility; associated coeliac disease, pancreatic exocrine insufficiency, SIBO; drugs (especially acarbose, metformin)
AIDS	Drug side-effects; lymphoma; opportunistic infections (cryptosporidiosis, CMV, HSV, Mycobacterium avium complex)
Institutionalised / hospitalised	<i>C. difficile</i> colitis; drug side-effects; faecal impaction with overflow diarrhoea; ischaemic colitis; tube feeding

Teach with **Main p. 83** · Chronic diarrhoea Source **IM p. 59**

C5-23

Malabsorption and coeliac disease Main pp. 84–92

◆ ADDITION

Maldigestion versus malabsorption

- ◆ **Maldigestion** is defective intraluminal hydrolysis of nutrients; **malabsorption** is defective mucosal absorption. Their presentations and complications are similar.
- ◆ Many malabsorptive disorders, including coeliac disease, present **subtly** or mainly with **extra-intestinal** features rather than diarrhoea or steatorrhoea. Malabsorption may even be only a laboratory finding.

Teach with Main p. 84 · Malabsorption Source IM p. 60

C5-24

◆ ADDITION

Coeliac disease: why each extra-intestinal feature occurs

Selected manifestations and probable causes (IM p. 66)

Manifestation	Probable cause
Dermatitis herpetiformis	Epidermal (type 3) tTG autoimmunity
Ecchymoses, petechiae	Vitamin K deficiency; rarely thrombocytopenia
Oedema	Hypoproteinaemia
Thrombocytosis, Howell–Jolly bodies	Hyposplenism
Secondary hyperparathyroidism	Calcium and/or vitamin D malabsorption with hypocalcaemia
Osteopenia, osteomalacia, osteoporosis	Ca/vitamin D malabsorption, secondary hyperparathyroidism, chronic inflammation
Elevated liver tests	Lymphocytic hepatitis
Peripheral neuropathy	Vitamin B12 and thiamine deficiency; immune-based neurological dysfunction
Ataxia	Cerebellar and posterior column damage
Short stature, delayed puberty, amenorrhoea, infertility	Malnutrition; hypothalamic–pituitary dysfunction
Weakness	Generalised muscle atrophy, hypokalaemia

Teach with Main pp. 89–91 · Coeliac disease Source IM p. 66

C5-30

◆ ADDITION

Coeliac disease: wider associations and later risks

- ◆ **Further associated conditions** (besides type 1 diabetes and autoimmune liver and thyroid disease): Addison disease, autoimmune haemolytic anaemia, immune thrombocytopenic purpura, **IgA deficiency**, **Down syndrome**, cystic fibrosis, Sjögren syndrome, **microscopic colitis**, epilepsy with cerebral calcification, sarcoidosis, recurrent pericarditis, iridocyclitis/choroiditis, some lung diseases (fibrosing alveolitis, idiopathic pulmonary haemosiderosis, bird-fancier's lung, cavitary lung disease) and macroamylasaemia.
- ◆ **Increased morbidity after diagnosis**: malignancy (non-Hodgkin lymphoma, **small-intestinal adenocarcinoma**); cardiovascular (ischaemic heart disease, cerebrovascular disease, atrial fibrillation, dilated cardiomyopathy); respiratory (COPD, asthma, tuberculosis, influenza).

Teach with **Main p. 90** · Coeliac disease (Fig 5.5) Source **IM p. 66**

C5-31

Diagnostic tests for malabsorption Main pp. 93–94

◆ ADDITION

Interpreting the general tests of absorption

- ◆ **Quantitative stool fat**: a borderline 8–14 g/24 h can occur in secretory or osmotic diarrhoea **not** caused by malabsorption. False negative if fat intake is inadequate; false positive with mineral-oil laxatives or rectal suppositories (e.g. cocoa butter).
- ◆ **Acid steatocrit**: a cheap, easy screening test for fat malabsorption; normal < 10%.
- ◆ **D-xylose test** (used in your book's algorithm, Fig 5.9): tests small-intestinal **mucosal** absorption, so it separates mucosal malabsorption from pancreatic insufficiency. Oral 25 g D-xylose; normal is > 4 g excreted in the urine over 5 h. False positives with renal failure, ascites or incomplete urine collection; it may also be abnormal in bacterial overgrowth and normal in mild or limited mucosal disease.
- ◆ **Lactose H₂ breath test**: a late peak (3–6 h) of > 20 ppm suggests lactose malabsorption.
- ◆ **Faecal elastase-1**: as sensitive as the secretin test for moderate-to-severe pancreatic insufficiency, but unreliable in mild insufficiency; false positives with high stool volume and mucosal disease. The **secretin test** is sensitive but labour-intensive and invasive.

Teach with **Main pp. 93–94** · Diagnostic tests Source **IM pp. 68–70**

C5-34/36

◆ ADDITION

Diagnosing bacterial overgrowth

- ◆ **Gold standard:** quantitative culture of jejunal aspirate; $> 10^5$ CFU/mL suggests overgrowth. False negatives occur with focal diverticula or overgrowth distal to the aspiration site.
- ◆ **Glucose H₂ breath test:** an early peak (within 2 h) of > 20 ppm suggests overgrowth. Contrast this with the late peak of lactose malabsorption. False negatives occur with non-hydrogen-producing organisms.
- ◆ **¹⁴C-D-xylose breath test:** sensitivity and specificity $> 90\%$, and more specific than the H₂ breath test. A non-radioactive ¹³C version is suitable for children and pregnant women.

Teach with Main p. 93 · Tests for bacterial overgrowth Source IM pp. 69–70

C5-35

◆ ADDITION

Biopsy and ileal-function tests

- ◆ **Small-bowel biopsy:** distal duodenal specimens are usually adequate; take several (4–5). Biopsy is **diagnostic** in intestinal infection, Whipple disease, abetalipoproteinaemia, agammaglobulinaemia, lymphangiectasia, lymphoma and amyloidosis. In coeliac disease and tropical sprue the findings are characteristic, but the diagnosis rests on **improvement after treatment**.
- ◆ **Permeability studies** are gaining favour for screening for small-intestinal disease and monitoring the response to treatment.
- ◆ **SeHCAT:** 7-day retention $< 10\%$ is abnormal and indicates bile acid malabsorption; liver disease and bacterial overgrowth can give false results.
- ◆ The **Schilling test** is no longer used.

Teach with Main p. 93 · Tests for mucosal disease and ileal function Source IM p. 69

C5-37

▲ DISCREPANCY · REVIEW

Stool fat and lactose breath test protocols

Main book	p. 93 (following IM's own text on p. 68): 70–100 g fat diet with stool collected over 2–3 days ; lactose dose 25 g .
GIT Internal Med	The test tables on pp. 69–70 give a 100 g fat diet with stool collected over 3 days , and a lactose dose of 1 g/kg body weight.
For review	IM's text and tables disagree with each other; the main book follows the text.

Teach with Main p. 93 · Diagnostic tests Source IM pp. 68–70

C5-38

06

CHAPTER 06 · TEACH AFTER MAIN BOOK FROM P. 95

Constipation

Internal Medicine 7 entries

Clinical assessment and testing

 Main pp. 96–100

● CLINICAL PEARL

History and examination clues

- ◆ In **acute-onset** constipation, always ask whether the patient can **pass gas**. Failure to pass gas can signal bowel obstruction.
- ◆ **Obstipation** is severe or complete constipation with practically no stool and **no flatus**. It may come with abdominal pain, nausea and vomiting, and points to obstruction.
- ◆ **DRE while bearing down**: pelvic floor dyssynergia is suggested when the anal sphincter **fails to relax** or the perineum **does not descend**.

Teach with Main pp. 97–98 · Clinical assessment Source IM p. 71

C6-03/04

◆ ADDITION

What anorectal testing measures

- ◆ **Anorectal manometry**: a pressure-sensitive catheter in the anorectum measures **resting and squeeze pressures** of the anal canal. It assesses the internal and external anal sphincters, **rectal sensation** and expulsion patterns.
- ◆ Defecography can also be performed with **MRI** (MRI defecography).

Teach with Main pp. 98, 100 · Diagnostic evaluation Source IM p. 72

C6-08

Disorders of gut–brain interaction and IBS

 Main pp. 101–105

◆ ADDITION

IBS: extra points

- ◆ **Functional chest pain of presumed oesophageal origin** is another oesophageal DGBI: chronic, recurrent pain or discomfort referred to the retrosternum.
- ◆ **Autonomic dysfunction** is also listed among the possible contributors to IBS.
- ◆ **History**: fibromyalgia is one of the associations found in IBS patients.
- ◆ **Limited laboratory tests**: besides faecal calprotectin and coeliac serology, IM lists **CBC** (anaemia or infection), **ESR and CRP** (inflammation) and **TSH** (thyroid dysfunction).

Teach with Main pp. 101–104 · DGBI; IBS Source IM pp. 73–75

C6-10/11/13/15

● CLINICAL PEARL

Why probiotics are not used in IBS

Probiotics are not prescribed in IBS because of a **lack of evidence supporting their efficacy**. This is the reason behind the main book's "no routine recommendation".

Teach with **Main p. 105** · Management of IBS Source **IM p. 75**

C6-18

▲ DISCREPANCY · REVIEW

IBS diagnostic criteria: which version?

The two sources quote different editions of the Rome criteria, so their pain-frequency thresholds differ. The 2-of-3 associated features (related to defecation, change in stool frequency, change in stool form) are the same in both.

Main book	p. 102 (Rome V): pain on average at least 3 days per month over the last 3 months.
GIT Internal Med	p. 75 (Rome IV): recurrent abdominal pain at least 1 day per week in the last 3 months.

Teach with **Main p. 102** · Diagnosis of IBS Source **IM p. 75**

C6-16

▲ DISCREPANCY · REVIEW

Low-FODMAP diet: which foods to limit?

The example lists of foods to limit do not match.

Main book	p. 104 lists foods high in fructans and oligosaccharides: onions, garlic, wheat, rye and legumes .
GIT Internal Med	p. 75 lists potatoes, brown rice, oats and almonds as examples of foods the patient should limit.

Teach with **Main p. 104** · Management of IBS Source **IM p. 75**

C6-17

▲ DISCREPANCY · REVIEW

IBS-C: the second drug option

Both sources give lubiprostone (chloride channel activator) as the first option but differ on the second.

Main book	p. 105 gives prucalopride (selective 5-HT ₄ agonist, prokinetic). Linaclotide appears only in the laxative table on p. 99, for refractory constipation.
GIT Internal Med	p. 75 gives linaclotide , a guanylate cyclase activator that softens stool and stimulates bowel movement.

Teach with **Main p. 105** · Management of IBS-C Source **IM p. 75**

C6-20

07

CHAPTER 07 · TEACH AFTER MAIN BOOK FROM P. 106

Foodborn Infection

Microbiology · Internal Medicine 0 entries

No substantive additions identified. GIT Internal Med has no food-borne infection chapter. Its infective-diarrhoea material (the exposure–pathogen table and *C. difficile* colitis) is given under Chapter 5 of this companion, next to the acute and refractory diarrhoea sections of your book where it is taught.

08

CHAPTER 08 · TEACH AFTER MAIN BOOK FROM P. 115

Inflammatory Bowel Disease

Internal Medicine · Pharmacology 8 entries

Pathology and diagnosis Main pp. 116–119

◆ ADDITION

Extra pathological and endoscopic clues

- ◆ In IM's comparison table the **rectum is always involved in UC** (only commonly in Crohn's disease).
- ◆ On radiology, **"collar-button" ulcers** are common in UC and only occasional in Crohn's disease.
- ◆ **Colonoscopy (your book's "Endoscopy" heading has no text):**
 - UC: friable mucosa, extensive ulceration and exudates.
 - Crohn's disease: a **single aphthous ulcer** is the earliest finding; multiple oedematous inflammatory polyps give the **cobblestone** appearance.

Teach with Main pp. 117–118 · UC vs Crohn's (Fig 8.1); Endoscopy Source IM pp. 76, 78

C8-02/08

▲ DISCREPANCY · REVIEW

Granulomas in ulcerative colitis

Main book

Fig 8.1 (p. 117) says granulomas **should not be present** in UC (microgranulomas may accompany ruptured crypt abscesses).

GIT Internal Med

The p. 76 pathology table lists granulomas in UC as **"occasional"**. IM's own p. 77 table agrees with the main book.

Teach with Main p. 117 · UC vs Crohn's (Fig 8.1) Source IM pp. 76–77

C8-03

◆ ADDITION

More conditions that mimic IBD

- ◆ Other colitides: **radiation colitis**, **Behçet's colitis**, **schistosomiasis**, and sexually transmitted colitis (gonococcus, *Chlamydia*, herpes, trauma).
- ◆ Bacterial colitis also includes **Shigella** and **invasive E. coli**.
- ◆ Crohn's look-alikes: **lymphoma**, **Yersinia** and **tuberculosis**.

Teach with Main p. 119 · Differential diagnosis (Fig 8.3) Source IM p. 79

C8-11

Treatment Main pp. 119–120

◆ ADDITION

The missing drug groups and a severity ladder

IM lists six drug groups for IBD. Two of them, **5-aminosalicylates (5-ASA/sulfasalazine)** and **probiotics/prebiotics**, do not appear in the main book's IBD chapter. IM also steps treatment up with severity:

Stepwise IBD therapy (IM p. 81)

Severity	Treatment
Mild	5-aminosalicylates / sulfasalazine; local steroid enemas; local 5-ASA enemas and suppositories
Moderate	Prednisolone; budesonide; immunosuppressant
Severe	Biological therapy; surgery

Teach with Main pp. 119–120 · Treatment Source IM pp. 80–81

C8-13/23

◆ ADDITION

5-ASA and budesonide: the formulation decides where the drug acts

Choose the formulation by disease site. Most act in the colon; Pentasa is released from the duodenum onwards; enemas reach the rectum and sigmoid, and suppositories the rectum.

5-aminosalicylates and budesonide (IM p. 80 drug table)

Drug	Dose	Release site
Sulfasalazine (Azulfidine)	2–6 g/day	Colon
Mesalamine (Asacol, Lialda, Apriso)	2.4–4.8 g/day	Distal ileum, colon
Olsalazine (Dipentum)	1–3 g/day	Colon
Balsalazide (Colazal)	6.25 g/day	Colon
Mesalamine (Pentasa)	2–4 g/day	Duodenum, jejunum, ileum, colon
Mesalamine (Rowasa) enema; suppository	4 g/day (enema); 1 g/day (suppository)	Rectum/sigmoid; rectum
Mesalamine (Canasa) suppository	1 g/day	Rectum
Budesonide (Entocort EC)	Induction 9 mg PO daily; maintenance 6 mg PO daily	Small intestine
Budesonide (MMX, UCERIS)	Induction 9 mg PO daily	Colon

Teach with Main pp. 119–120 · Treatment; corticosteroids Source IM p. 80

C8-14/15

▲ DISCREPANCY · REVIEW

Systemic steroid doses

Main book

pp. 119–120 give methylprednisolone **10–30 mg IV q6–8h** and prednisone **40–60 mg/day** (IM's own p. 80 text says the same).

GIT Internal Med

The p. 80 drug table gives methylprednisolone **40–60 mg IV daily** and prednisone **0.25–0.75 mg/kg PO daily**.

Teach with **Main pp. 119–120** · Corticosteroids Source **IM p. 80**

C8-17

◆ ADDITION

Immunomodulators and biologics: full list and doses

Your book names azathioprine, methotrexate, cyclosporin, infliximab and vedolizumab without doses. IM adds **6-mercaptopurine** and four more biologics: **adalimumab**, **golimumab**, **certolizumab pegol**, **natalizumab**.

Immunomodulators and biologics (IM p. 80 drug table)

Drug	Induction / dose	Maintenance
6-Mercaptopurine	1.5 mg/kg/day	–
Azathioprine	2.5 mg/kg/day	–
Methotrexate	25 mg SC weekly × 4 months	15–25 mg SC weekly
Cyclosporine	2–4 mg/kg/day IV	–
Infliximab	5 mg/kg IV weeks 0, 2, 6	5–10 mg/kg IV every 8 weeks
Adalimumab	160 mg SC week 0, 80 mg week 2	40 mg SC every other week
Golimumab	200 mg SC week 0, 100 mg week 2	100 mg SC every 4 weeks
Certolizumab pegol	400 mg SC weeks 0, 2, 4	400 mg SC every 4 weeks
Natalizumab	300 mg IV every 4 weeks	–
Vedolizumab	300 mg IV at 0, 2 and 6 weeks	300 mg IV every 8 weeks

Teach with **Main p. 120** · Immunomodulators; biologic therapy Source **IM p. 80**

C8-20/21

Complications Main p. 121

● CLINICAL PEARL

Complications grouped by disease

- ◆ **UC:** toxic megacolon and malignancy.
- ◆ **Crohn's disease:** abscesses and fistulas, obstruction, and perianal disease.

Teach with **Main p. 121** · Complications Source **IM p. 82**

C8-25

09

CHAPTER 09 · TEACH AFTER MAIN BOOK FROM P. 122

Parasitic GI Diseases

Parasitology 0 entries

No substantive additions identified. GIT Internal Med has no parasitology chapter. Its passing references to *Giardia*, amoebiasis and hookworm are covered under Chapters 3 and 5 of this companion or are already in your book.

10

CHAPTER 10 · TEACH AFTER MAIN BOOK FROM P. 140

Small Intestine and Colon Surgery

Surgery 0 entries

No substantive additions identified. GIT Internal Med contains no surgical material on the small intestine or colon (your Chapter 10 covers Meckel's diverticulum).

11

CHAPTER 11 · TEACH AFTER MAIN BOOK FROM P. 145

Diverticulosis of the Colon

Surgery · Internal Medicine 0 entries

No substantive additions identified. GIT Internal Med mentions diverticular disease only as a cause of lower GI bleeding and left iliac fossa pain. Those points are already in your book or appear in Chapter 3 of this companion (lower GI bleeding).

12

CHAPTER 12 · TEACH AFTER MAIN BOOK FROM P. 151

Appendix

Surgery 0 entries

No substantive additions identified. GIT Internal Med has no appendix section; appendicitis appears only in its list of right iliac fossa pain causes, which your book already gives.

13

CHAPTER 13 · TEACH AFTER MAIN BOOK FROM P. 162

Intestinal Obstruction

Surgery · Emergency / Acute Care 0 entries

No substantive additions identified. GIT Internal Med has no obstruction section. Its imaging points on obstruction (plain films, CT in high-grade obstruction, enteroclysis) are given under Chapter 1 of this companion, where your book teaches imaging.

14

CHAPTER 14 · TEACH AFTER MAIN BOOK FROM P. 189

Polyposis of Colon

Surgery · Pathology 0 entries

No substantive additions identified. GIT Internal Med has no polyposis section. Its one polyp-related point, that a polyp found at colonoscopy should not be assumed to explain chronic blood loss, is given under Chapter 3 (chronic GI bleeding).

15

CHAPTER 15 · TEACH AFTER MAIN BOOK FROM P. 195

Colorectal Cancer

Surgery · Pathology 0 entries

No substantive additions identified for main-book pp. 195–197, the part of Chapter 15 inside this companion's scope (the supplied book file ends at p. 197). GIT Internal Med has no colorectal cancer section. Its related points (CT colonography technique, unprepared CT in frail patients, right-colon cancer in chronic blood loss) are given under Chapters 1 and 3.

Source for every entry: GIT Internal Med (page shown on each entry). Compared against the YMnotes Digestive System study edition, Version 3, printed pages 1–197. Entries marked Discrepancy · Review are shown for the teacher to decide and are not resolved here.

رَبِّ زِدْنِي عِلْمًا

[سورة طه: 114]



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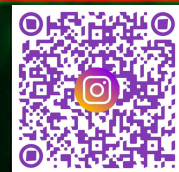
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